

Simons Dissertation Fellowship Proposal

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Introduction

My research revolves around *derived categories* in algebraic geometry. Broadly speaking, the derived category of a space X can be considered a more flexible version of the category of vector bundles/coherent sheaves on X , but it should also be thought of as a “higher invariant” [7], capturing data such as Hochschild cohomology, K-theory, and some Hodge theory. My reasons for interest in the derived category are the following:

- The existence of *semiorthogonal decompositions*, which describe how one derived category is assembled from smaller pieces.
- The existence of interesting derived equivalences between varieties, as in the study of *Fourier-Mukai transforms*, and derived equivalences between varieties and noncommutative algebras (like quiver algebras), as in the study of *tilting bundles* and generalizations.
- The objects in a derived category having good moduli theory, with the notion of moduli spaces of *Bridgeland stable* objects, where the spaces vary as a *stability condition* crosses walls in a certain complex manifold.

In particular, I like the following program: first understand a derived category by producing semiorthogonal decompositions or equivalences to “easier” triangulated categories; and next, use this understanding to produce t-structures on your derived category, which one hopes can be refined to Bridgeland stability conditions.

As far as the immediate future goes, I am currently working on a project with my advisor Aaron Bertram, and my peers Jonathon Fleck and Liebo Pan, about applying Bridgeland stability to find generalizations (for surfaces) of certain theorems controlling the syzygies of a curve embedded by an adjoint divisor.

Preparation

In preparation for this program, I have spent the much of my time following the semiorthogonal decompositions of well-known spaces. Starting with Beilinson’s seminal work [3] producing a full strong exceptional collection on $\mathbb{P}^n$ (the best type of semiorthogonal decomposition), I then moved onto generalizations of Beilinson’s argument as in Kapranov’s [6] use of Koszul duality [4] and Borel-Weil-Bott from representation theory to construct exceptional

collections on quadrics. In turn, Kuznetsov [10] repeated Kapranov’s argument in families to produce semiorthogonal decompositions for quadric fibrations. Continuing in this direction, I want to better understand semiorthogonal decompositions created by birational geometry as pioneered by Orlov [11] and applied to great success for example in [13, 12] producing semiorthogonal decompositions for moduli spaces of vector bundles on curves.

With the structure of these derived categories better understood, I have also tracked how to produce t-structures and their refinements, Bridgeland stability conditions. In particular, projective space and quadrics, both having full strong exceptional collections, admit derived equivalences to quivers via tilting theory [5, 9]. Transferring the standard heart of a quiver through the equivalence, one can then adapt the theory of GIT quiver moduli [8] as in [1] to produce so-called *algebraic* Bridgeland stability conditions. This technique has been helpful for producing Bridgeland stability conditions, especially on higher dimensional varieties. On the other hand, Bertram and others have produced so-called geometric stability conditions on *all* surfaces by using torsion pairs and the Bogomolov inequality [2]. The project I am working on with Bertram and my peers Fleck and Pan uses these stability conditions.

Research Plan

todo

Career Goals

I am very much enjoying academia as a PhD candidate, with respect to research, organizing seminars, and being an instructor. I plan to continue onwards to become a postdoc and eventually a professor.

References

- [1] Daniele Arcara, Aaron Bertram, Izzet Coskun, and Jack Huizenga. The minimal model program for the Hilbert scheme of points on $\mathbb{P}^2$ and bridgeland stability. (arXiv:1203.0316), March 2012. arXiv:1203.0316 [math].
- [2] Daniele Arcara, Aaron Bertram, and Max Lieblich. Bridgeland-stable moduli spaces for k-trivial surfaces. (arXiv:0708.2247), August 2007. arXiv:0708.2247 [math].
- [3] A. A. Beilinson. Coherent sheaves on $\mathbb{P}^n$ and problems of linear algebra. *Functional Analysis and Its Applications*, 12(3):214–216, 1978.
- [4] Alexander Beilinson, Victor Ginzburg, and Wolfgang Soergel. Koszul duality patterns in representation theory. *Journal of the American Mathematical Society*, 9(2):473–527, 1996.
- [5] Alastair Craw. Explicit methods for derived categories of sheaves.
- [6] M. M. Kapranov. On the derived categories of coherent sheaves on some homogeneous spaces. *Inventiones Mathematicae*, 92(3):479–508, October 1988.
- [7] Bernhard Keller. *Derived categories and tilting*, page 49–104. Cambridge University Press, 1 edition, January 2007.

- [8] A. D. King. Moduli of representations of finite dimensional algebras. *The Quarterly Journal of Mathematics*, 45(4):515–530, 1994.
- [9] Alastair King. Tilting bundles on some rational surfaces. unpublished manuscript.
- [10] Alexander Kuznetsov. Derived categories of quadric fibrations and intersections of quadrics. (arXiv:math/0510670), October 2005. arXiv:math/0510670.
- [11] D O Orlov. Projective bundles, monoidal transformations, and derived categories of coherent sheaves. *Russian Academy of Sciences. Izvestiya Mathematics*, 41(1):133–141, February 1993.
- [12] Jenia Tevelev. Braid and phantom. (arXiv:2304.01825), December 2023. arXiv:2304.01825 [math].
- [13] Jenia Tevelev and Sebastián Torres. The BGMN conjecture via stable pairs. (arXiv:2108.11951), November 2023. arXiv:2108.11951 [math].